In the stack shown in figure 7.1, the blockchain client node is called *web3 provider* because it hosts (and thus provides) the classes and functions specified in web3. The bottom layer is the actual blockchain functions, including the smart contract execution environment and the ledger recording the blocks of Txs. The Ganache test chain you used in chapters 4 and 6, and an Ethereum node implemented in the Go language (geth node), are examples of a blockchain client node.

In all, figure 7.1 provides an overview of the architecture of a typical Dapp stack. You'll use this stack as a guideline to implement a blockchain feature called a micropayment channel for addressing the global problem of plastics cleanup.

Before you launch into designing the Dapp, let's learn a little bit about web3. If you prefer, you can skip to application design in section 7.2.

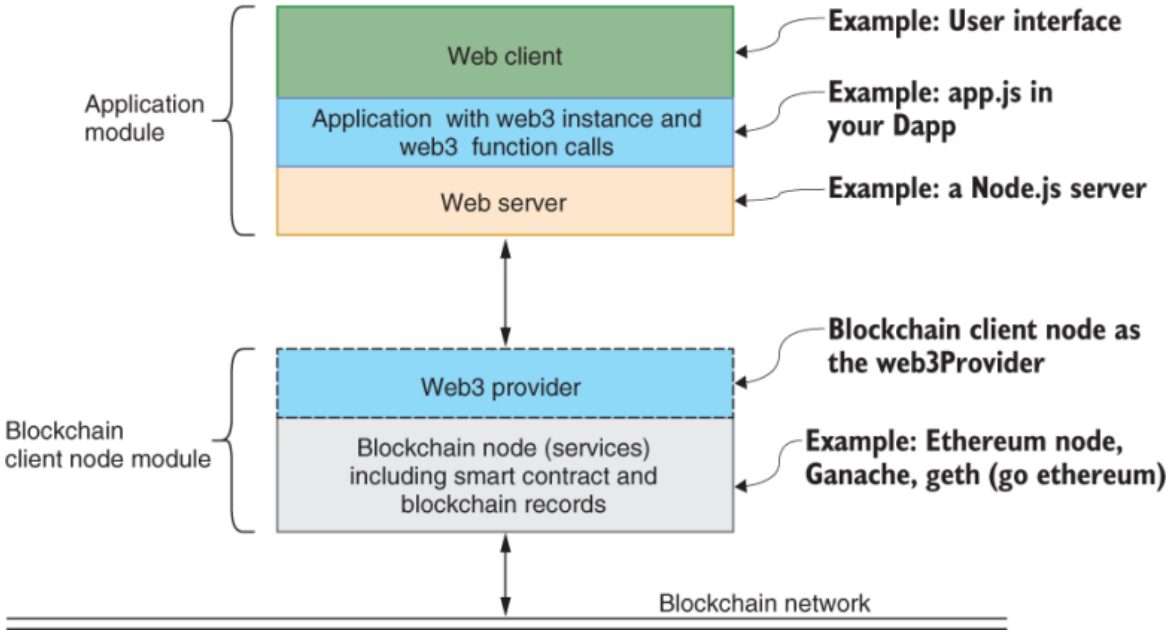


Figure 7.1 Role of web3 in the blockchain-based Dapp stack